

WASP-43b

R-1907

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_190304 · created 2026-08-05T23:19:27+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458546.8546 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.036
Transit depth (Rp/Rs)^2	3.74 +/- 1.41 [%]
Orbital Inclination (inc)	79.78 +/- 2.22
Airmass coefficient 1 (a1)	0.997 +/- 0.013
Airmass coefficient 2 (a2)	0.001 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.33 %
Best Comparison Star	#3 - [353, 26]
Optimal Aperture	4.62
Optimal Annulus	13.1
Transit Duration (day)	0.042 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.193 ± 0.036 vs 0.1594 (deviation 0.93σ)
Duration fit vs published	0.042 d vs 0.0512 d (deviation 0.71σ)
Mid-time O–C	-4.2 min
Depth SNR	5.4
Residual scatter	1.33 %

Light curve

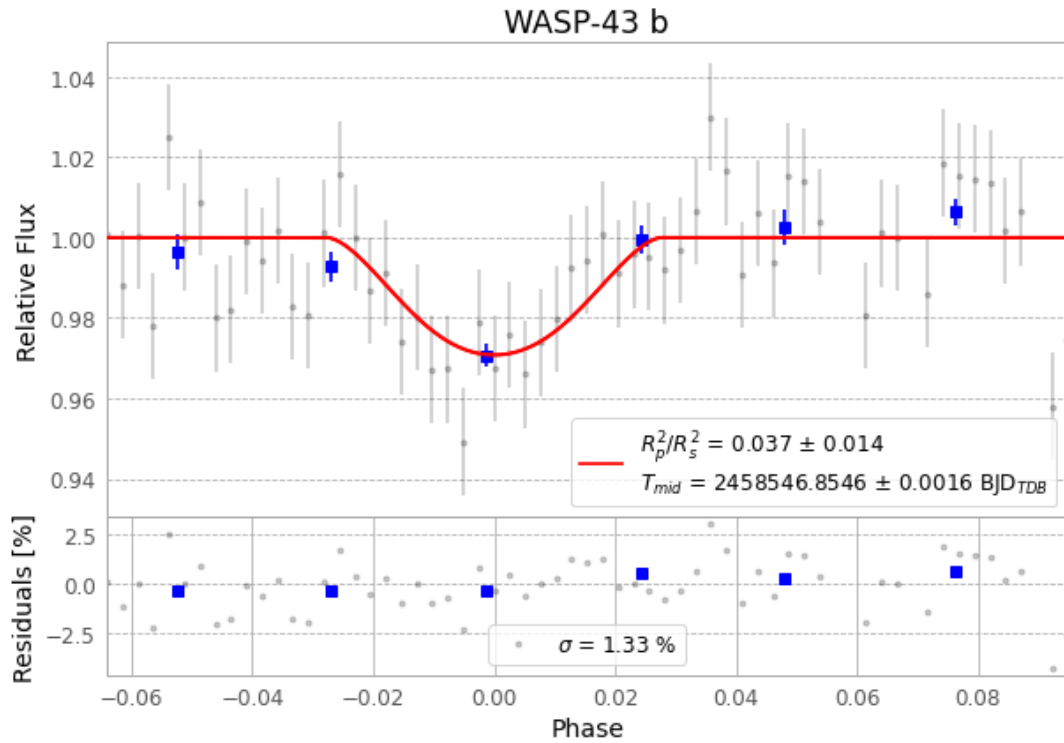


Figure 1 — FinalLightCurve_WASP-43 b_2019-03-04.png (SHA-256 6f051d56d22b2b34...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [430, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 61396f1a9f4b3ea7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), WASP-43190304070612.FITS → WASP-43190304101512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-190304.log (a7e816bb27a7...), FinalLightCurve_WASP-43 b_2019-03-04.png (6f051d56d22b...), FinalLightCurve_WASP-43 b_2019-03-04.csv (994b0947432b...)

Compute resources

Wall time 132.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-05 23:19Z.